

Mayan Saravanabavan

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EDUCATION

University of Toronto, BAsC in Electrical Engineering (Co-op)

Expected Apr 2029

SKILLS

Languages: Python, C, C++, Verilog, RISC-V Assembly, HTML, CSS

Technologies: Git, Quartus Prime, ModelSim, LTspice, Altium Designer

Lab Equipment: Oscilloscope, Function Generator, Digital Multimeter, Power Supply, Soldering Iron

EXPERIENCE

Undergraduate Research Assistant, CANUE

July 2025 – Present

- Developed a Python web application enabling researchers to obtain custom geospatial health and climate data
- Designed a Streamlit interface to upload a CSV file of locations and dates, select datasets, and view outputs
- Implemented dynamic PostgreSQL query logic using psycopg2 to retrieve datasets based on user input
- Maintained a PostgreSQL database, regularly adding new data and indexing it to improve query performance
- Contributed to a research paper that describes the purpose, functionality, and usage of the application

Team Member, UTWind

Sept 2024 – Present

- Researched buck converter components to ensure reliable operation in a small-scale wind turbine
- Tested the buck converter circuit using LTspice simulations and multimeter measurements

PROJECTS

Power Amplifier and Filter for HF Radio Transceiver

- Designed the final transmit stage of a team-built radio, amplifying and filtering signals for antenna broadcast
- Simulated the full signal chain in LTspice, achieving 9.6 W output power with 1.23% THD at 14 MHz
- Captured schematics and completed PCB layout in Altium Designer within a 50.8 mm × 90.2 mm board envelope
- Assembled the PCB and validated output power and distortion against design targets using an oscilloscope

Redesign Proposal for Chestnut Residence Building

- Evaluated HVAC, lighting, and window systems to diagnose energy inefficiencies across 2 building floors
- Proposed floor-level energy-efficient design improvements aligned with building performance goals
- Validated proposed designs using building energy modelling software, achieving a 2.3% overall energy reduction

DE1-SoC Arcade Game

- Developed a Verilog-based asteroid-shooting game with real-time rendering on a 640×480 VGA display
- Configured PS/2 keyboard input handling to control ship movement via WASD and bullet firing via spacebar
- Integrated bounding-box collision detection to identify ship-asteroid overlaps and trigger a game-over state

Resume Builder

- Launched an HTML, CSS, and JavaScript web application that generates custom resumes
- Implemented LaTeX file generation to produce professionally formatted resumes from user input

Document Scanner

- Developed a C++ document scanning application using OpenCV to automatically detect documents in images
- Integrated Tesseract OCR to extract and store text from processed document images in under 2 seconds